

When we do so, we should get the following output, which represents a tree with a depth of two:

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```
[['netusoft', -inf, 4.0, [['hhmb', -inf, 4.0, [['agea', -inf, 15.0,
8.035714285714286], ['agea', 15.0, inf, 6.997666564322997]]], ['hhmb', 4.0, inf,
[['eduyrs', -inf, 11.0, 7.263969171483622], ['eduyrs', 11.0, inf, 8.0]]]],
['netusoft', 4.0, inf, [['hhmb', -inf, 1.0, [['agea', -inf, 66.0,
7.135361428970136], ['agea', 66.0, inf, 7.621993127147766]]], ['hhmb', 1.0, inf,
[['rlgdgr', -inf, 5.0, 7.743893678160919], ['rlgdgr', 5.0, inf,
7.9873320537428025]]]]]
```

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*Listing 9-5: A representation of a decision tree using nested lists*

What you see here is a collection of lists nested within each other. These nested lists represent our full decision tree, though it's not as easy to read as [Figure 9-1](#). In each level of nesting, we find a variable name and its range, just like we saw with the simple tree illustrated in [Figure 9-3](#). The first level of nesting shows us the same branch we found in [Figure 9-3](#): a branch that represents respondents whose value of `netusoft` was less than or equal to 4.0. The next list, nested within the first, begins with `hhmb, -inf, 4.0`. This is another branch of our decision tree that branches from the branch we just examined, and consists of people whose self-reported household size is 4 or less. If we drew the portion of a decision tree that we've looked at in our nested list so far, it would look like [Figure 9-4](#).

We can continue to look at the nested lists to fill in more branches of our decision tree. Lists that are nested within other lists correspond to branches that are lower on the tree. A nested list branches from the list that contains it. The terminal nodes, instead of containing more nested lists, have an estimated happiness score.